

Jianing(Jenny) Chen

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EDUCATION

University of California, San Diego, San Diego, CA
B.S. in Bioengineering, Minor in Mathematics (GPA: 3.95)

Sep. 2022 – June. 2026

Honors & Awards: Phi Beta Kappa, Provost Honors

Relevant courses: Data Science in Practice, Mathematical Statistics, Data Analysis and Inference, Modeling & Computation in Bioengineering, MATLAB Programming for Engineers

SKILLS

- **Programming:** Python, SQL, R, Java
- **Data Analysis & Machine Learning:** pandas, NumPy, scikit-learn, statistical inference, regression, classification, model evaluation
- **Data Visualization:** Tableau, Power BI, matplotlib, seaborn, Excel
- **Engineering & Lab Tools:** MATLAB, ImageJ, OriginLab, SolidWorks, Fusion 360, Git

RESEARCH EXPERIENCE

Bioengineering Senior Design (Swallow Diagnostics), UC San Diego

Summer 2025 – Present

- Built an automated **MATLAB pipeline** to process raw high-resolution impedance manometry (HRIM) signals into structured quantitative datasets
- Extracted swallow-level features using **ManoView** and computed biomechanical metrics of esophageal function (mechanical work, distension index, stiffness)
- Generated patient-level visualizations and performed statistical analyses (**t-tests, ANOVA**) to evaluate treatment outcomes across patients
- Collaborated with faculty mentors and peers to synthesize findings into poster presentations, showcasing both technical analysis and patient-care implications; Results will be presented at the **2026 Bioengineering Day and Digestive Disease Week (DDW)**

Tai-Seale Laboratory, Research Assistant, UC San Diego

June. 2025 – Present

- Cleaned and preprocessed large-scale healthcare communication datasets using **Python (pandas, NumPy, scikit-learn)**, ensuring accuracy and usability for analysis
- Performed **exploratory data analysis (EDA)** to characterize emotional expression and communication patterns
- Evaluated multiple **NLP and machine learning models (e.g., VADER, RoBERTa)** and their output variability on clinical text
- Visualized sentiment score distributions using **matplotlib and seaborn** to compare emotional patterns between **AI-generated and human-written responses**
- Summarized findings in technical reports and visual dashboards, presenting results to cross-disciplinary teams to guide decision-making on AI-assisted health communication tools

PROFESSIONAL EXPERIENCE

Shanghai Institute of Biological Products Co., Ltd., Data Science Intern, Shanghai

June. 2024 – Sep. 2024

- Collected, cleaned, and analyzed production and quality control datasets from 500+ Varicella vaccine batches using Excel, SQL, and **Python (pandas, NumPy)** to identify variability patterns and improve supply consistency
- Designed and automated QC reports that tracked defect rates, contamination checks, and compliance metrics, improving issue detection efficiency by 10%
- Applied data visualization tools (**Tableau/Power BI**) to monitor vaccine yield, defect rates, and process KPIs, enabling management to make data-driven decisions
- Collaborated with cross-functional teams (QA, manufacturing, and biosafety) to document workflows and integrate real-time data capture systems for improved traceability

PROJECT EXPERIENCE

Breast Cancer Surgery Outcome Analysis, UC San Diego

Mar. 2025 – June. 2025

- Cleaned and structured the METABRIC clinical dataset (~1,900 samples) using **Python (pandas, NumPy)**, applying cohort segmentation techniques and addressing missing values to prepare the dataset for stratified analysis
- Performed **exploratory data analysis (EDA)** to map relationships between surgical success rates and breast cancer subtypes, producing statistical summaries and correlation heatmaps
- Developed visualizations with **seaborn and matplotlib** that highlighted risk patterns across subgroups, enabling faculty reviewers to assess the strength of subtype-specific hypotheses

Comparative Analysis of Classical Machine Learning Models on Biomedical Datasets, UC San Diego

Sep. 2025 – Jan. 2026

- Compared **five classical machine learning classifiers** (Logistic Regression, SVM, Random Forest, KNN, Naive Bayes) across **four biomedical datasets** with varying size and feature structure
- Designed a unified evaluation framework using **multiple train-test splits** (20/80, 50/50, 80/20) and systematic hyperparameter tuning in **scikit-learn**
- Analyzed how dataset characteristics (**dimensionality, noise, feature correlation**) affected model performance using **accuracy metrics, visualization, and statistical testing**

LEADERSHIP EXPERIENCE

International Services & Engagement Office (ISEO), Project Intern, UC San Diego

Sep. 2025 – Jan. 2026

- Organized 10+ cross-cultural campus events, engaging 500+ participants and increasing participation in international student programming
- Secured campus funding and coordinated logistics with partner organizations to support event planning and outreach